


# The University of California, Santa Cruz

## PHYS 133: Intermediate Laboratory

## Lab 1: Atomic Spectra

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| *Author* | *Lab Partner* |
|---|---|
| Zach Warner | Arsh Sayyad |

*Instructor*

Dr. Pablo Saz Parkinson

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# Atomic Spectra

Zach Warner*[1]

[1]*Department of Physics, University of California, Santa Cruz*

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**Abstract**

In this experiment, we used a visible light spectrometer to first obtain an accurate value for the spacing between the lines of the diffraction grating used in the spectrometer, then use that information to attempt to derive the Rydberg constant of the hydrogen atom. By measuring the known wavelength spectral lines of a mercury lamp, we calculated that the grating spacing was an average of 3880 ± 90.4 nm between each line in the grating. Using this, we attempted to determine the Rydberg constant with a hydrogen lamp, however, we unfortunately collected unusable data due to a misinterpretation of the lab manual, and we were unable to collect new data, so we were not able to calculate a value for the Rydberg constant. After this, we used the spectrometer to compare the spectral lines of a helium-neon laser to those of helium and neon lamps to better understand which energy level transitions give the laser its distinct red color. We measured the average angle between the laser's first order spectral line and the nearest helium spectral line to be 0.576° ± 0.008° for the left spectral lines and 0.567° ± 0.008° for the right ones, which is a significant separation. Then, we measured the average angle between the same spectral line of the laser and the nearest spectral line of neon, which resulted in an angle of 0.050° ± 0.008° for the left and 0.083° ± 0.008° for the right, which almost put the two lines on top of each other, pointing to neon being the source of the laser's red color.

*ztwarner@ucsc.edu

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## 1 Introduction

The purpose of this experiment is to use a visible light spectrometer to verify the expected energy level pattern of the hydrogen atom and derive the Rydberg constant. The first part of this experiment involves using a mercury lamp as a known wavelength standard to calculate the spacing of the lines of the diffraction grating. Then, we can use the grating spacing with a hydrogen lamp, so we can measure the spacing of the spectral lines and use that to calculate the Rydberg constant. Finally, we will be comparing the spectral lines of a helium-neon laser to both helium and neon lamps to see if any of them line up in a meaningful way, pointing to the energy level transition happening in the laser.

The visible light spectrometer works by using a series of lenses and a diffraction grating to split the light from a source into its component spectral lines, which are spaced based on the different energy level transitions happening in the lamp [1]. Due to the diffraction grating, the spectral lines also repeat depending on the angle ($\theta_m$), grating spacing (d), and wavelength ($\lambda$) of the spectral lines according to this equation. (see Equation 1):

$$\frac{m\lambda}{d} = \sin\theta_m, \tag{1}$$

Where,

$$\theta_m = \frac{1}{2}(\alpha_1 + \alpha_2), \tag{2}$$

Alpha 1 and alpha 2 are the angles that the spectral lines were measured at on either side of the central line, and m is a positive integer.

$\alpha_1$ and $\alpha_2$ are calculated from the angles measured on the left and right vernier scales

($\phi_{1,m}$ and $\phi_{2,m}$ respectively) using the following formula. (see Equation 3):

$$\alpha_{1,2} = \frac{|\phi_{1,m} - \phi_{1,m=0}| + |\phi_{2,m} - \phi_{2,m=0}|}{2} \tag{3}$$

The wavelengths ($\lambda$) of the hydrogen spectrum are given by this equation (see Equation 4):

$$\frac{1}{\lambda} = R_H(\frac{1}{n_1^2} + \frac{1}{n_2^2}) \tag{4}$$

Where $R_H$ is the Rydberg constant and $n_1$ and $n_2$ are the principal quantum numbers of the initial and final energy states for the energy level transition responsible for the spectral line. However, the only energy level transitions that produce visible light all end with a principal quantum number of two. With this information, we can use our measurements of the angles of the spectral lines to calculate the Rydberg constant.

Figure 1 is an image of a visible light spectrometer [1].

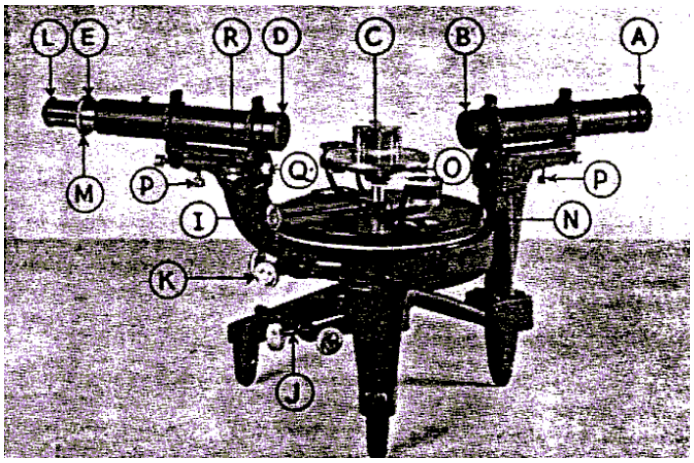



**Figure 1:** Taken from the Lab Manual [1].

## 2 Apparatus and Procedure

In setting up this experiment, the main piece of equipment involved in the visible light spectrometer, which consists of the collimator (responsible for redirecting the incoming light into parallel beams), an adjustable slit on the far end of the collimator (which filters out most of the light that is not coming in parallel), the diffraction grating (which diffracts the light into separate spectral lines), the telescope objective lens (which can be rotated around the diffraction grating to see the spectral lines), and an eyepiece (which we look through to see the spectral lines and line them up consistently using the cross-hairs inside).

Figure 2 is a schematic diagram of a visible light spectrometer.

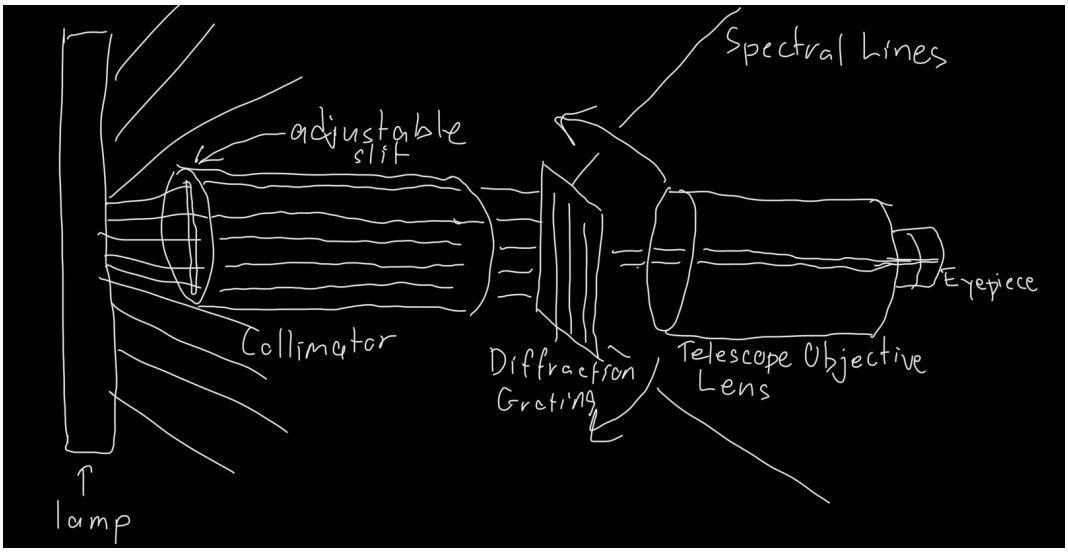



**Figure 2:** Schematic diagram drawn based on the description in the lab manual [1]

We began by positioning the far end of the spectrometer's collimator one centimeter away from a lamp, in which we placed a mercury Geissler tube. The Geissler tubes used in this experiment emit light only from the gas contained in each tube, which, for this experiment, are mercury, hydrogen, neon, and helium. Before we were able to start measuring spectral lines, we had to first calibrate the spectrometer, so that we can accurately measure the angles. First, we lined up the collimator and telescope objective lenses with no diffraction grating, so that we could see the light from the lamp through the eyepiece. Then, we adjusted the

collimator focus knob until the edges of the light from the lamp were clearly defined. Next, we adjusted the eyepiece focus until we could clearly see the cross-hairs on top of the light from the lamp. With the light and the cross-hairs clearly visible, we then placed a mirror in the diffraction grating holder (the grating table), so we could position the grating table perfectly pointed straight toward the telescope objective lens. With the mirror in place, we connected the eyepiece leads to a battery to shine a light from the cross-hairs at the mirror and back to the eyepiece. With the reflection of the cross-hairs visible through the eyepiece, we then adjusted the three leveling screws of the grating table and the grating table rotation clamp until the reflection of the cross-hairs lined up perfectly behind the actual cross-hairs in the eyepiece, meaning the table was pointing perfectly at the telescope objective lens.

With the grating table, collimator, and telescope objective lens aligned and in focus, we then disconnected the eyepiece leads from the battery and replaced the mirror with the diffraction grating. Then, we aligned the cross-hairs directly with the right edge of the central line and used the grating table rotation clamp and tangent screw to set the angle readings on either side of the spectrometer to 90 degrees on the left vernier scale and 270 degrees on the right.

With the spectrometer now calibrated and the angle of the central line set, we then started looking for the spectral lines for the mercury lamp. To do this, we released the rotation clamp on the telescope lens and rotated it to the right until we saw the first set of spectral lines (m=1). We then used the fine control screw to position the cross-hairs directly on the right edge of the green 546.074 nm line and took the measurements from both the left and right vernier scales. Then we repeated this process, but rotated the telescope lens to the left to measure the angle of the left first-order (m=1) green spectral line. We then repeated this process for the left and right second (m=2) and third-order (m=3) green spectral lines, moving the telescope lens further to the left and right respectively.

After this, we replaced the mercury Geissler tube with a Hydrogen one inn order to

measure the angles of its spectral lines. In taking this dara, however, we made a mistake and accidentally used the same procedure as for the mercury tube, which meant we did not have the data necessary to calculate the Rydberg constant. Unfortunately, by the time this error was realized, we could no longer use the spectrometer to collect the correct data. So, just like the mercury tube, we used the grating table rotation clamp and tangent screw to set the angle of the right edge of the central line to 90 degrees on the left vernier scale and 270 degrees on the right. Similar to the mercury lamp, we then used the telescope lens rotation clamp and fine control screw to align the cross-hairs with the right edge of the first-order (m=1) light blue (486.133 nm) spectral lines on both the left and right sides of the central line. We measured the angle on both the left and right vernier scale at both the left and right first-order spectral lines. We then repeated this process for the second (m=2) and third-order (m=3) spectral lines on both the left and right sides.

Finally, we set up a helium-neon laser to shine directly at the Geissler tube, in front of the spectrometer's adjustable slit. We then placed a piece of cardboard a hole cut in it at the height of the collimator slit. The piece of cardboard had a piece of paper covering the bottom half of the hole, which we placed directly in front of where the laser was shining on the Geissler tube. We adjusted the laser and piece of cardboard until the paper covered half of the view through the eyepiece and shined the laser onto the paper, so that we could see the central line of the laser in the top half of the eyepiece and the central line of the Geissler tube in the bottom half. Then, we replaced the hydrogen Geissler tube with a helium one and once again used the grating table rotation clamp and tangent screw to set the angle of right edge the central lines to 90 degrees on the left vernier scale and 270 degrees on the right. We then used the telescope lens rotation clamp and fine control screw to align the cross-hairs with the right edge of the right, fist-order spectral line of the laser and measured the angle from both the left and right vernier scales. We then repeated this process for the nearest right, first-order spectral line of the helium lamp and compared the angles. Then, we repeated this process again for the left spectral lines of both the helium-neon laser and

the helium Geissler tube. Next, we replaced the helium tube with a neon tube and ensured that the central lines were still set to an angle of 90 degrees on the left vernier scale and 270 degrees on the right. We then used the telescope lens rotation clamp and fine control screw to align the right edge of the right, first-order spectral line of the neon lamp nearest to that of the laser and measured the angle on both the left and right vernier scales. We then repeated this process for the neon lamp's left spectral lines.

# 3 Results

## 3.1 *Determination of the Grating Spacing*

Below are the measured left and right vernier scale angles ($\phi_1$ and $\phi_2$ respectively) of the left half of the green 546.074 nm spectral lines from the mercury Geissler tube and the values of $\alpha_1$, derived from $\phi_1$ and $\phi_2$ using Equation 3 (see Table 1).

**Table 1**
Measured angles of left spectral lines from the mercury Geissler tube.

| m | $LeftScale[\phi_1][^\circ]$ | $RightScale[\phi_2][^\circ]$ | $\alpha_1[^\circ]$ |
|---|---|---|---|
| 0 | 90.067 | 269.983 | $0 \pm 0.008$ |
| 1 | 80.833 | 260.650 | $9.284 \pm 0.008$ |
| 2 | 71.467 | 251.300 | $18.642 \pm 0.008$ |
| 3 | 61.117 | 241.000 | $28.967 \pm 0.008$ |

The error in every measurement of $\phi_1$ and $\phi_2$ from the vernier scales is $\pm$ 0.008°. The errors of $\alpha_1$ were calculated using Equation 9.

This table contains the measured left and right vernier scale angles ($\phi_1$ and $\phi_2$ respectively) of the right half of the green spectral lines from the mercury tube and the values of $\alpha_2$, derived from $\phi_1$ and $\phi_2$ using Equation 3 (see Table 2).

**Table 2**
Measured angles of right spectral lines from the mercury Geissler tube.

| m | $LeftScale[\phi_1][^\circ]$ | $RightScale[\phi_2][^\circ]$ | $\alpha_2[^\circ]$ |
|---|---|---|---|
| 0 | 90.067 | 269.983 | $0 \pm 0.008$ |
| 1 | 99.433 | 279.300 | $9.342 \pm 0.008$ |
| 2 | 109.217 | 289.083 | $19.125 \pm 0.008$ |
| 3 | 119.267 | 299.167 | $29.192 \pm 0.008$ |

The error in every measurement of $\phi_1$ and $\phi_2$ from the vernier scales is $\pm$ 0.008°. The errors of $\alpha_2$ were calculated using Equation 9.

Combining the results for $\alpha_1$ and $\alpha_2$ according to Equation 2 gave us these values for $\theta_m$ (see Table 3).

**Table 3**
Computed values of $\theta_m$ from $\alpha_1$ and $\alpha_2$.

| $\alpha_1[^\circ]$ | $\alpha_2[^\circ]$ | $\theta_m[^\circ]$ |
|---|---|---|
| 0 | 0 | 0 |
| 9.284 | 9.342 | $9.313 \pm 0.006$ |
| 18.642 | 19.125 | $18.884 \pm 0.006$ |
| 28.967 | 29.192 | $29.080 \pm 0.006$ |

The uncertainty in every $\alpha_1$ and $\alpha_2$ is $\pm$ 0.008°, as derived from Equation 9. The uncertainty in $\theta_m$ was calculated using Equation 12.

Then, with the known value of the wavelength ($\lambda$=546.074 nm), we used the values of $\theta_m$ and m and Equation 1 to calculate a grating spacing (d) for each m, which we averaged to get a final result for the grating spacing of the diffraction grating. The corresponding values of m, $\theta_m$, and d are shown in the table below (see Table 4).

**Table 4**
Computed values of d from $\theta_m$ and m.

| $m$ | $\theta_m[^\circ]$ | $d[nm]$ |
|---|---|---|
| 1 | 9.313 | $4896 \pm 262$ |
| 2 | 18.884 | $3374 \pm 59.2$ |
| 3 | 29.080 | $3370 \pm 36.4$ |
| | | $3880 \pm 90.4$ |

The uncertainty in every $\theta_m$ is $\pm$ 0.006°. The uncertainty in the grating spacing for each m was calculated from Equation 14, while the uncertainty in the average grating spacing was calculated using Equation 16.

The final result we calculated for the grating spacing was therefore 3880 nm $\pm$ ().

### 3.2 *Determination of the Rydberg Constant*

With the grating spacing calculated, we then moved on to determining the Rydberg constant using the spectral lines from a hydrogen Geissler tube. However, when we were gathering our data, we accidentally used the same procedure as for the mercury tube, which meant all of our data was for different modes (m) of the same wavelength. This meant that we could not use Equation 4, because it requires different wavelengths of the same mode. So, just like with the mercury tube, the table below contains the measured left and right vernier scale angles ($\phi_1$ and $\phi_2$ respectively) of the left half of the blue 486.133 nm spectral lines from the hydrogen Geissler tube (see Table 5). This data cannot be used to calculate the Rydberg constant, but I will include it anyway, as it is what we measured.

**Table 5**
Measured angles of left spectral lines from the hydrogen Geissler tube.

| m | $LeftScale[\phi_1][^\circ]$ | $RightScale[\phi_2][^\circ]$ |
|---|---|---|
| 0 | 90.067 | 269.983 |
| 1 | 81.783 | 261.617 |
| 2 | 73.300 | 253.150 |
| 3 | 64.417 | 244.300 |

The error in every measurement of $\phi_1$ and $\phi_2$ from the vernier scales is $\pm$ 0.008°.

Similarly, this table contains the measured left and right vernier scale angles ($\phi_1$ and $\phi_2$ respectively) of the right half of the blue 486.133 nm spectral lines from the hydrogen

Geissler tube (see Table 6).

**Table 6**
Measured angles of right spectral lines from the hydrogen Geissler tube.

| m | $LeftScale[\phi_1][^\circ]$ | $RightScale[\phi_2][^\circ]$ |
|---|---|---|
| 0 | 90.067 | 269.983 |
| 1 | 98.333 | 278.217 |
| 2 | 106.800 | 286.717 |
| 3 | 115.600 | 295.500 |

The error in every measurement of $\phi_1$ and $\phi_2$ from the vernier scales is $\pm$ 0.008°.

### 3.3 *The Spectra of Neon, and of a Helium-neon Laser*

Below is a table containing the measured left and right vernier scale angles ($\phi_1$ and $\phi_2$ respectively) of the left half of the first-order spectral lines from the helium-neon laser, as well as the nearest spectral lines of the helium and neon Geissler tubes (see Table 7). For this experiment, all of the readings for the left and right vernier scales at the central line were the same (89.983° $\pm$ 0.008° and 269.967° $\pm$ 0.008°), as the laser never moved.

**Table 7**
Measured angles of left spectral lines from the helium-neon laser, helium Geissler tube, and neon Geissler tube.

| Working Gas | $LeftScale[\phi_1][^\circ]$ | $RightScale[\phi_2][^\circ]$ |
|---|---|---|
| Helium-neon laser | 79.117 | 259.017 |
| Helium | 78.500 | 258.483 |
| Neon | 79.067 | 258.967 |

The error in every measurement of $\phi_1$ and $\phi_2$ from the vernier scales is $\pm$ 0.008°.

The following table contains the measured left and right vernier scale angles ($\phi_1$ and $\phi_2$ respectively) of the right half of the first-order spectral lines from the helium-neon laser, as well as the nearest spectral lines of the helium and neon Geissler tubes (see Table 8).

**Table 8**
Measured angles of right spectral lines from the helium-neon laser, helium Geissler tube, and neon Geissler tube.

| Working Gas | $LeftScale[\phi_1][^\circ]$ | $RightScale[\phi_2][^\circ]$ |
|---|---|---|
| Helium-neon laser | 100.850 | 280.750 |
| Helium | 101.417 | 281.317 |
| Neon | 100.950 | 280.816 |

The error in every measurement of $\phi_1$ and $\phi_2$ from the vernier scales is $\pm$ 0.008°.

As is evident from the tables, the nearest spectral lines of helium are slightly off when compared to the helium-neon laser, being off by an average of 0.576° $\pm$ 0.008° on the left and 0.567° $\pm$ 0.008° on the right. However, the nearest spectral lines of neon are very close to those of the helium-neon laser, off by an average of 0.050° $\pm$ 0.008° on the left and 0.083° $\pm$ 0.008° on the right. These averages are taken by adding the difference in the left vernier scale reading of the target gas and the laser to the difference in the right vernier scale reading of both and dividing by two. Their errors were calculated using Equation 19.

## 4 Discussion and Conclusion

Overall, the results we ended up with are very similar to the expected results. The grating spacing we calculated ended up being 3880 ± 90.4 nm, and the grating spacing we used was labeled with, "300 lines per mm," which corresponds to a grating spacing of about 3333 nm. This puts our measurement at about 16.5% greater than the expected result. However, it is important to note that the actual grating spacing may be slightly off from the expected value due to the manufacturing process [1]. It is also important to note that the two values of the grating spacing before averaging with the smallest error were significantly closer to the expected value, at 3374 ± 59.2 nm and 3370 ± 36.4 nm for the second and third-order spectral lines respectively.

Unfortunately, as stated before, we were not able to use our data to calculate the Rydberg constant. In the helium-neon laser experiment, we expected the spectral line of the laser to line up almost exactly with one of the spectral lines in both the helium lamp and neon lamp. However, as we saw, the spectral line of the laser was off from the nearest spectral line of Helium by an average of 0.576° ± 0.008° on the left and 0.567° ± 0.008° on the right, which was a very noticeable difference through the spectrometer eyepiece. The nearest spectral line of neon was significantly closer, though, at an average of 0.050° ± 0.008° on the left and 0.083° ± 0.008° on the right, which made the lines overlap some through the spectrometer eyepiece. This leads me to believe that the laser's red color comes primarily from the neon, with helium not influencing the color nearly as much.

## 5 Acknowledgements

My lab partner, Arsh Sayyad, helped me gather some of the data for these experiments. Dr. Pablo Saz Parkinson and his teaching assistant, Swosti Choudhury, also provided some guidance throughout the experiments.

## 6 Appendix: Error Analysis

The errors in the values of $\alpha_1$ and $\alpha_2$ were calculated using Equation 9.

$$\frac{\delta\alpha_{1,2}}{\delta\phi_{1,m}} = \frac{1}{2} \tag{5}$$

$$\frac{\delta\alpha_{1,2}}{\delta\phi_{1,m=0}} = -\frac{1}{2} \tag{6}$$

$$\frac{\delta\alpha_{1,2}}{\delta\phi_{2,m}} = \frac{1}{2} \tag{7}$$

$$\frac{\delta\alpha_{1,2}}{\delta\phi_{2,m=0}} = -\frac{1}{2} \tag{8}$$

$$\sigma_{\alpha_{1,2}} = ((\frac{\delta\alpha_{1,2}}{\delta\phi_{1,m}})^2\sigma^2_{\phi_{1,m}} + (\frac{\delta\alpha_{1,2}}{\delta\phi_{1,m=0}})^2\sigma^2_{\phi_{1,m=0}} + (\frac{\delta\alpha_{1,2}}{\delta\phi_{2,m}})^2\sigma^2_{\phi_{2,m}} + (\frac{\delta\alpha_{1,2}}{\delta\phi_{2,m=0}})^2\sigma^2_{\phi_{2,m=0}})^{\frac{1}{2}} \tag{9}$$

The error in $\theta_m$ was calculated using Equation 12.

$$\frac{\delta\theta_m}{\delta\alpha_1} = \frac{1}{2} \tag{10}$$

$$\frac{\delta\theta_m}{\delta\alpha_2} = \frac{1}{2} \tag{11}$$

$$\sigma_{\theta_m} = ((\frac{\delta\theta_m}{\delta\alpha_1})^2\sigma^2_{\alpha_1} + (\frac{\delta\theta_m}{\delta\alpha_2})^2\sigma^2_{\alpha_2})^{\frac{1}{2}} \tag{12}$$

The error in the grating spacing (d) was calculated using Equation 14.

$$\frac{\delta d}{\delta\theta_m} = -m\lambda\cot\theta_m\csc\theta_m \tag{13}$$

$$\sigma_d = \left|\frac{\delta d}{\delta\theta_m}\right|\sigma_{\theta_m} \tag{14}$$

The error in the average grating spacing ($d_{avg}$) was calculated according to Equation 16.

$$\frac{\delta d_{avg}}{\delta d_1} = \frac{\delta d_{avg}}{\delta d_2} = \frac{\delta d_{avg}}{\delta d_3} = \frac{1}{3} \tag{15}$$

$$\sigma_{d_{avg}} = ((\frac{\delta d_{avg}}{\delta d_1})^2 \sigma_{d_1}^2 + (\frac{\delta d_{avg}}{\delta d_2})^2 \sigma_{d_2}^2 + (\frac{\delta d_{avg}}{\delta d_3})^2 \sigma_{d_3}^2)^{\frac{1}{2}} \tag{16}$$

The error in the average distances between spectral lines of the helium-neon laser and the helium and neon Geissler tubes ($\theta_{gas*,avg}$) was calculated using Equation 19.

*gas refers to the gas in the Geissler tube the laser is being compared to.

$$\frac{\delta \theta_{gas,avg}}{\delta \phi_{1,gas}} = \frac{\delta ltheta_{gas,avg}}{\delta \phi_{2,gas}} = \frac{1}{2} \tag{17}$$

$$\frac{\delta \theta_{gas,avg}}{\delta \phi_{1,laser}} = \frac{\delta ltheta_{gas,avg}}{\delta \phi_{2,laser}} = -\frac{1}{2} \tag{18}$$

$$\sigma_{\theta gas,avg} = ((\frac{\delta \theta_{gas,avg}}{\delta \phi_{1,gas}})^2 \sigma_{\phi_{1,gas}}^2 + (\frac{\delta \theta_{gas,avg}}{\delta \phi_{1,laser}})^2 \sigma_{\phi_{1,laser}}^2 + (\frac{\delta \theta_{gas,avg}}{\delta \phi_{2,gas}})^2 \sigma_{\phi_{2,gas}}^2 + (\frac{\delta \theta_{gas,avg}}{\delta \phi_{2,laser}})^2 \sigma_{\phi_{2,laser}}^2)^{\frac{1}{2}} \tag{19}$$

# References

[1] G. Brown. *Laboratory Manual: Physics 133*. 2024.